



Module 1: Welcome to AWS Academy Natural Language Processing

AWS Academy Natural
Language Processing

Module overview

Sections

1. Course overview
2. What is NLP?
3. Business problems solved by using NLP
4. NLP roles
5. What's new with AWS

Module objectives

At the end of this module, you should be able to:

- Identify course prerequisites and objectives
- Describe natural language processing (NLP)
- Describe business problems that are solved by using NLP
- Describe various roles that require NLP
- Identify NLP tasks for each role
- Explore new technologies at AWS



Section 1: Course Objectives and Overview

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Course prerequisites

- IT technical knowledge
- Completion of AWS Machine Learning Foundations
- Intermediate skills with Python programming
- General knowledge of applied statistics
- Familiarity with machine learning (ML) concepts
- Working knowledge of distributed systems
- Familiarity with general networking concepts
- Working knowledge of multi-tier architectures

Course objectives

- Describe NLP
- Understand how to apply the ML pipeline to NLP
- Implement text extraction to obtain data from webpages
- Create a solution for a sentiment analysis business problem
- Use Amazon Comprehend to implement a topic modeling solution
- Build a solution that uses AWS services to transcribe and translate text from multi-media
- Build a solution using a combination of algorithms and Amazon Machine Learning services

Course outline

- Module 1: Welcome to AWS Academy NLP
- Module 2: Introduction to NLP
- Module 3: Processing Text for NLP
- Module 4: Implementing Sentiment Analysis
- Module 5: Introducing Information Extraction
- Module 6: Introducing Topic Modeling
- Module 7: Working with Languages
- Module 8: Course Wrap-Up

Module 2: Introduction to NLP

Module sections:

- NLP and machine learning
- Common NLP tasks
- Walkthrough of an NLP problem
- Evolution of NLP



Module 3: Processing Text for NLP

Module sections:

- Text processing overview
- Extracting text
- Tokenizing text
- Common processing steps
- Advanced processing
- Storing and visualizing unstructured data



Module 4: Implementing Sentiment Analysis

Module sections:

- Scenario introduction
- Identifying text processing steps
- Examining algorithms for sentiment analysis
- Challenge lab: discussion and walkthrough



Module 5: Introducing Information Extraction

Module sections:

- Information extraction overview
- Types of information extraction



Module 6: Introducing Topic Modeling

Module sections:

- Introduction to topic modeling
- Identifying the approach
- Introduction to Amazon Comprehend
- Algorithms
- Making recommendations based on text



Module 7: Working with Languages

Module sections:

- NLP and languages
- Identifying languages
- Translating text with Amazon Translate



Module 8: Course Wrap-Up

Module sections:

- Capstone project
- Course recap
- Bridging to certification





Section 2: What is NLP?

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NLP example

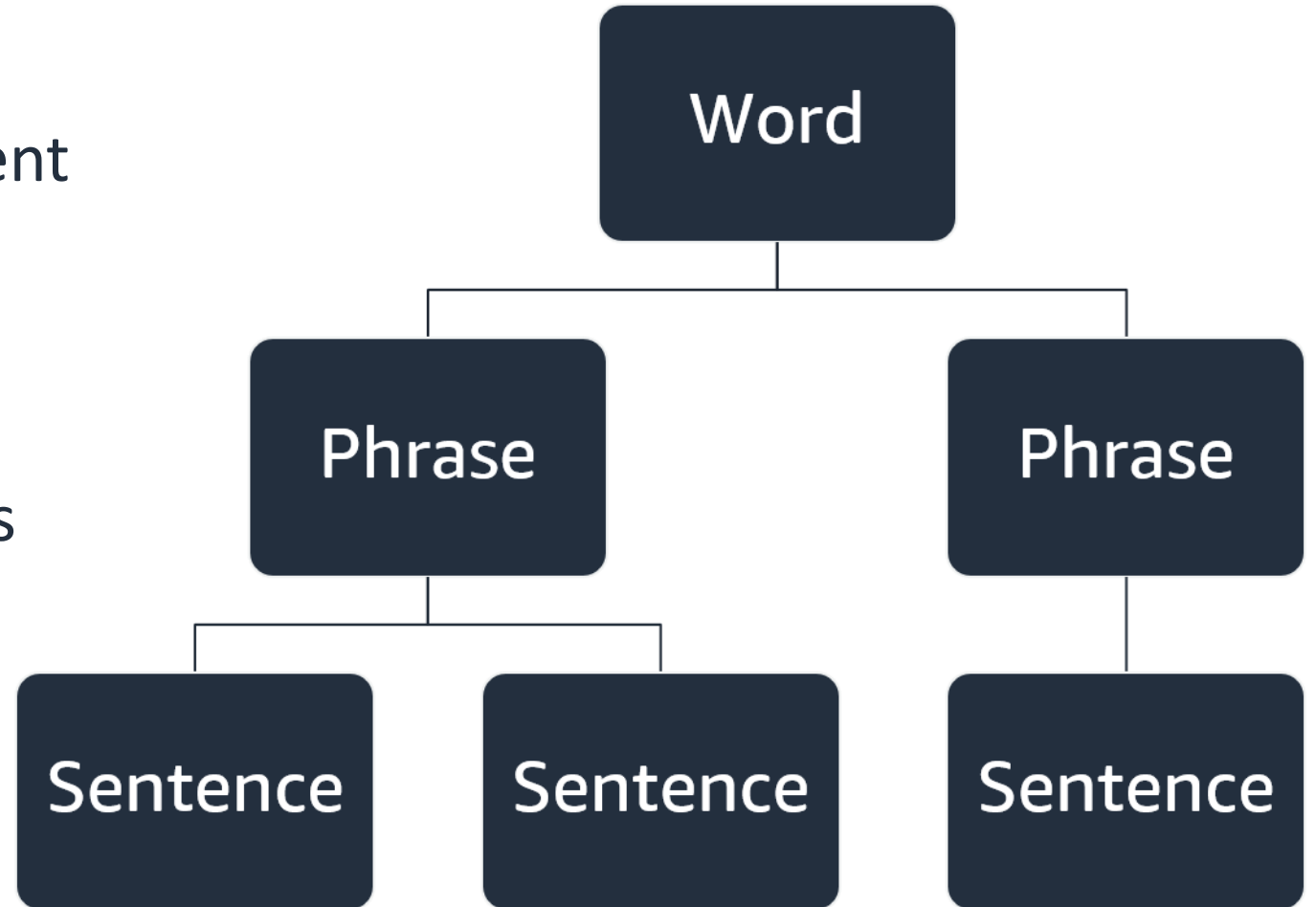
“Alexa, what’s it like outside?”



What is NLP?

NLP develops computational algorithms to analyze and represent human language automatically.

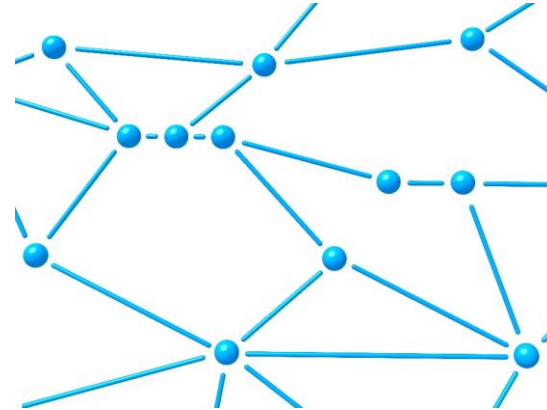
By evaluating the structure of language, ML systems can process large sets of words, phrases, and sentences.



NLP challenges



Lack of precision



Many complex dependencies



Meaning that is based on context



Lack of structure



Section 3: Business problems solved by using NLP

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Natural language processing use cases



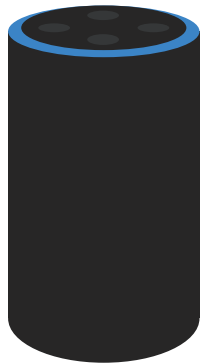
Search applications



Translations



Market and social research



Human machine
interfaces



Content management



Chatbots

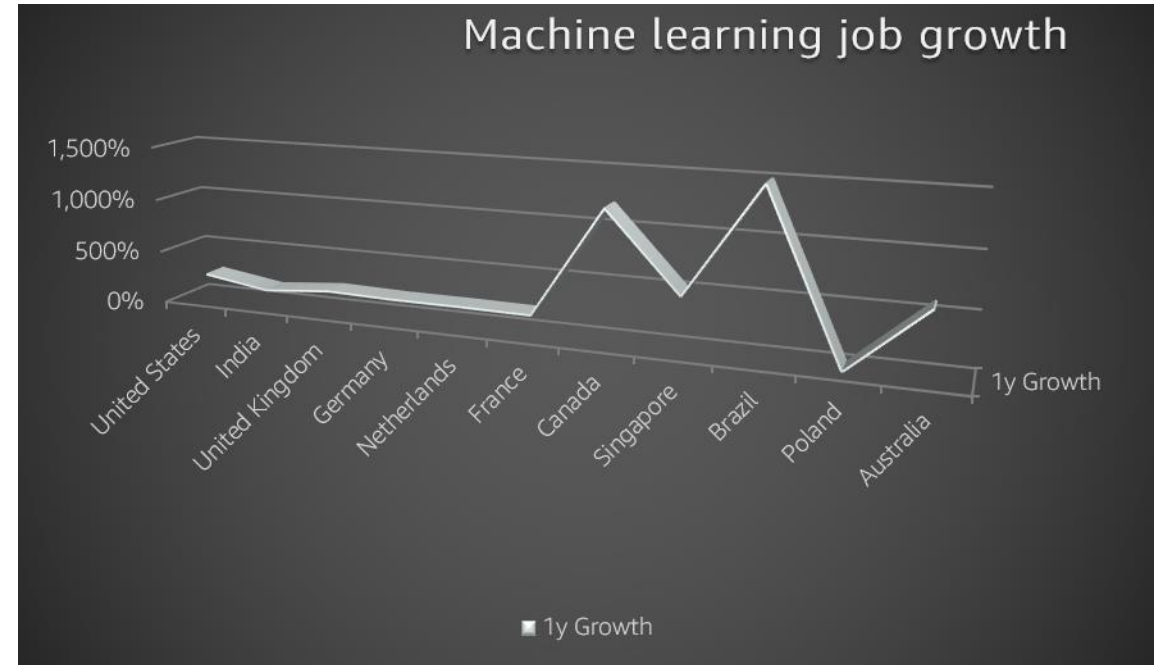


Section 4: NLP roles

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Careers in ML

- Demand for machine learning (ML) skills is very high and growing
- Variety of ML roles work on NLP problems
- Technical skills with the fastest growth in demand for ML roles
 - Python programming
 - SQL
 - Data science
 - Data mining
 - Data analysis



Data scientist or data analyst

- Applying knowledge of statistics and analytical skills to interpret data
- Data scientists often have degrees in statistics, computer science, or economics
- Some programming skills are required
- Typical tasks: Acquire and prepare data, exploratory data analysis, present data analysis



ML engineer

- Emphasis on programming and system design skills
- Often have background as a developer or software architect
- Some knowledge of statistics required
- Typical tasks: Evaluating and selecting algorithms, training and evaluating models, designing NLP applications



Applied science researcher

- Applies machine learning technology to a specific domain
- Requires knowledge of both the domain and machine learning
- Typical tasks: Using ML services to conduct research, applying NLP tools to a domain-specific application



Developer

- Integrates ML with software applications
- Requires strong application development skills and machine learning knowledge
- Typical tasks: Incorporating NLP models or services into an application



Activity: NLP jobs scavenger hunt

- Find a job description for one of the four NLP roles outlined in this module
- Good search locations include:
 - [Indeed](#)
 - [Glassdoor](#)
 - [Monster](#)
- Be prepared to present your findings to the class



Section 5: What's new at AWS

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What is Amazon CodeWhisperer?



CodeWhisperer

AI-powered code generator for IDEs and code editors

- AI coding companion:
 - Generates code suggestions based on comments and existing code
 - Offers real-time support for code authoring directly within your integrated development environment (IDE)
- AI security scanner:
 - Helps identify hard-to-find vulnerabilities
 - References multiple standards and best practices

Content processed by CodeWhisperer Professional is not stored or used for service improvement.

Code generation

- Code suggestions
- Code completion
- Code generation from comments
- Alternate code suggestions
- Option to accept or reject
- Reference tracking for code that resembles open-source training data

```
1 import boto3
```

```
1 import boto3  
2 # create an s3 bucket named cw95323
```

< 1/3 > Accept tab Accept Word ⌘ → ...

```
2 # create an s3 bucket named cw95323  
  s3 = boto3.resource('s3')  
  s3.create_bucket(Bucket='cw59323')  
  # upload a file to the bucket
```

Security scan

```
1 import boto3
2
3 def upload_file(bucket_name, file_path):
4     s3 = boto3.client('s3')
5     with open(file_path, 'rb') as file:
6         s3.upload_fileobj(file, bucket_name, file_path)
7         print("File uploaded.")
8
9 bucket_name = input("Enter bucket name: ")
10 file_path = input("Enter file path to upload: ")
11 upload_file(bucket_name, file_path)
```

PROBLEMS



CWE-22 – Path traversal: Constructing path names with unsanitized user input can lead to...

Benefits of Amazon CodeWhisperer



Value to developers

- Increase velocity.
- Spend less time writing code.
- Receive help directly within your IDE.
- Find security vulnerabilities in your code.



Value to organizations

- Use at all experience levels.
- Support open-source attribution.
- Reduce the risk of security vulnerabilities.
- Increase code quality and developer productivity.



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.